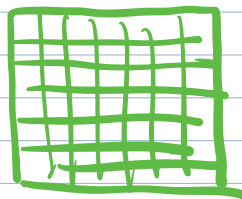


A GENTLE START

$\mathcal{X} \equiv$ Domain Set (Inputs)

Example: Digit Recognition.

$x \in \mathcal{X}$ is an $d \times d$ image.



$\mathcal{Y} \equiv$ label set. $\begin{cases} \mathcal{Y} \text{ DISCRETE} \\ \mathcal{Y} \text{ CONTINUOUS} \end{cases}$

TODAY: $\mathcal{Y} = \{0, 1\}$

BINARY CLASSIFICATION
PROBLEM.

$S \equiv$ Training set

$\{(x_i, y_i), i = 1 \dots m\}$ $x_i \in \mathcal{X}$
 $y_i \in \mathcal{Y}$

LEARNING ALGORITHM

HYPOTHESIS
CLASS.

$$A(S) \mapsto h \in \mathcal{H}$$

$$\text{s.t. } h : \begin{array}{ccc} x & \mapsto & y \\ \uparrow & & \uparrow \\ \mathcal{X} & & \mathcal{Y} \end{array}$$

DATA GENERATION ASSUMPTION

Simple: $x \in \mathcal{X} \quad x \sim \mathcal{D}_x$

x_i i.i.d.

$\exists f$ [UNKNOWN LABELING
FUNCTION]

$$f: x \mapsto y$$

$$\text{s.t. } \forall x_i \quad f(x_i) = y_i$$

MORE REALISTIC

$$(x, y) \sim \mathcal{D}$$



$$x \sim \mathcal{D}_x$$

$$y|x \sim \mathcal{D}_{y|x}$$

$$p(x, y) = p(y|x)p(x)$$

ERROR FUNCTION

$$z = (x, y)$$

Loss

$$l(h, z)$$



MEASURE HOW GOOD IS THE FUNCTION h TO DESCRIBE THE DATA POINT z

$$z = (x, y)$$

$$h(x) \approx y$$



FOR BINARY CLASSIFICATION

$$y = \{0, 1\}$$

A VERY COMMON LOSS FUNCTION IS THE SO-CALLED 0-1 LOSS.

$$l(h, z) = \begin{cases} 1 & \text{if } h(x) \neq y \\ 0 & \text{if } h(x) = y \end{cases}$$

(Risk)

[TRUE RISK]

[GENERALIZATION ERROR]

$$L_D(h) = \mathbb{E}_D[l(h, z)]$$

WARNING: IN THE "SIMPLE" CASE $x \sim D_x$, $y = f(x)$

$D_{y|x}$ IS A "DELTA MEASURE". IN ANY CASE

$$D = D_{y|x} D_x$$

↑ JOINT OVER (x, y)

COMPUTING

$\hat{h}_S$

"ESTIMATE"
OF h ("GOOD")
THAT CAN BE
COMPUTED ONLY USING
DATA IN S

$$S = \{(x_i, y_i), i = 1 \dots m\}$$

$$\forall h \in \mathcal{H} \quad \ell(h, z_i)$$

WE WOULD LIKE $\ell(h, z_i)$ SMALL
 $\forall i \in 1 \dots m$

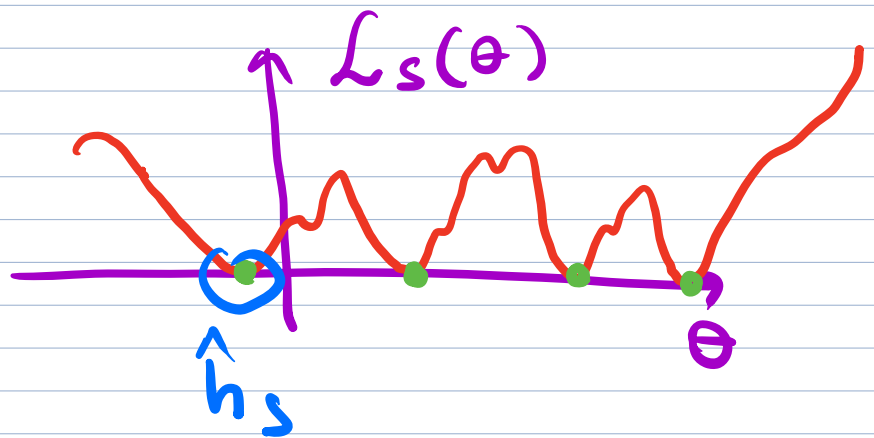
$$\underbrace{L_S(h)} = \frac{1}{m} \sum_{i=1}^m \ell(h, z_i)$$

↳ EMPIRICAL RISK
(TRAINING ERROR)

$$\hat{h}_S \in \arg \min_{h \in \mathcal{H}} L_S(h)$$

EMPIRICAL RISK MINIMIZATION

EXAMPLE: $h \sim \theta \in \mathbb{R}$



QUESTION: IS $L_D(\hat{h}_S)$
SMALL OR NOT?

GUESS:

$\hat{h}_S$ minimizes $L_S(h)$

$$L_S(\hat{h}_S) = \frac{1}{m} \sum_{i=1}^m \ell(\hat{h}_S, z_i) \approx L_D(\hat{h}_S)$$

↑
CAREFUL !!!

IT MAY HAPPEN THAT

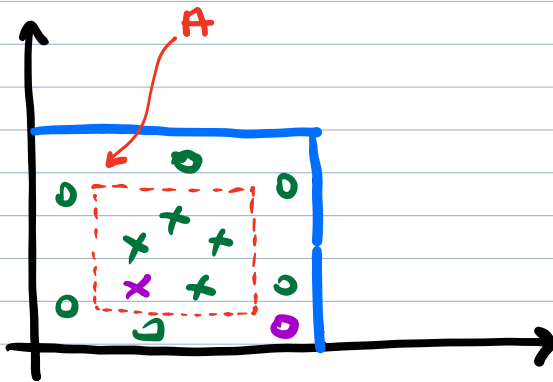
$$L_D(\hat{h}_S) \gg L_S(\hat{h}_S)$$

THE DIFFERENCE

$L_D(\hat{h}_S) - L_S(\hat{h}_S)$ IS CALLED

GENERALIZATION GAP

EXAMPLE:



$$X = [0,1] \times [0,1]$$

TRUE UNKNOWN LABELING
FUNCTION:

$$f(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

$x = "1"$

$0 = "0"$

PROBLEM: find $h \in \mathcal{H}$ s.t.

$$h(x) = y$$

$$\hat{h}_S \in \underset{h \in \mathcal{H}}{\operatorname{argmin}} \quad \frac{1}{m} \sum_{i=1}^m \ell(h, z_i)$$

IF $\mathcal{H}$ IS THE SET OF ALL POSSIBLE
FUNCTIONS THAT MAP $X \mapsto \{0,1\}$

IF WE TAKE

$$\hat{h}_S(x) = \begin{cases} 1 & \text{if } \exists i \text{ s.t. } \begin{matrix} x = x_i \\ y_i = 1 \end{matrix} \\ 0 & \text{otherwise} \end{cases}$$

CLEARLY $\mathcal{L}_S(\hat{h}_S) = 0$

ASSUMPTION 1 $x_i \sim \mathcal{U}([0,1] \times [0,1])$

ASSUMPTION 2 $\text{AREA}(A) = 1/2$

$$\mathcal{L}_D(\hat{h}_S) = \mathbb{E}_D[l(\hat{h}_S, z)]$$

$$\begin{aligned} &= 1 \cdot \underbrace{\mathbb{P}_D[l(\hat{h}_S, z) = 1]}_{\text{AREA}(A)} + 0 \cdot \mathbb{P}[l(\hat{h}_S, z) = 0] \\ &= \mathbb{P}[x \in A] = 1/2 \end{aligned}$$

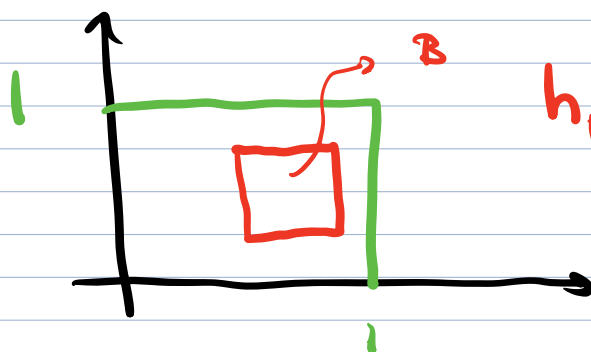


$$L_D(\hat{h}_S) = 1/2$$

$$L_S(\hat{h}_S) = 0$$

HOW TO CURE THIS PROBLEM?

$\mathcal{H} = \{ h(x) \text{ s.t. } h(x) \text{ IS THE INDICATOR FUNCTION OF ANY POSSIBLE SQUARE IN } [0,1] \times [0,1] \}$



$$h_B(x) = \begin{cases} 1 & x \in B \\ 0 & x \notin B \end{cases}$$

HOMEWORK!

WHAT WOULD

$\hat{h}_S(x)$ BE IN THIS CASE

IN THE NEXT LECTURE WE WILL
CONSIDER THE CASE IN WHICH

$$\mathcal{H} = \{h_1, \dots, h_m\}$$

$m = |\mathcal{H}|$ CARDINALITY
OF $\mathcal{H}$

WE WILL LINK m (COMPLEXITY)
WITH $m \equiv (\# \text{ OF TRAINING EXAMPLES})$

AND A MEASURE OF HOW BIG

$$L_D(\hat{h}_S) - L_S(\hat{h}_S) \text{ IS}$$